

Perfect Self-Containment and MDL

Why the Universe Has to Select Itself

UGP Physics Tutorial Series

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UGP Physics Tutorial Series

Prerequisites (read first): *The GTE Polynomial: A Step-by-Step Cheat Sheet · From the Arithmetic Substrate to Particle Masses*

Next tutorials: *How MDL Selects the 19-Bit Polynomial*

See also: *The Φ_{MDL} Field*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero **sorry** and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Puzzle

The central question

Why do the laws of physics have the specific values they do? Who — or what — chose them?

1.1 The mystery of the 25 dials

The Standard Model of particle physics is the most precise physical theory ever constructed. It predicts the magnetic moment of the electron to 12 decimal places, describes the strong force, the weak force, and electromagnetism, and accounts for essentially every subatomic observation ever made.

But it has a dirty secret: it comes with **25 free parameters** — 25 numerical dials that nobody knows how to set from first principles. The mass of the electron is just what it is. The number of particle generations (three) is just what it is. The fine-structure constant $\alpha \approx 1/137$ is just what it is. You measure them, you write them into the theory, and the theory works. But the theory cannot explain *why* those values rather than others.

This is a deep problem. Imagine an architect who designs a beautiful building but whose blueprint requires 25 arbitrary numbers — the ratio of ceiling height to door width, the exact colour of each brick — that have no internal reason. The building stands, but it was designed around coincidences, not principles. Our universe is like that building.

1.2 The three standard answers — and why none is fully satisfying

Answer 1: They are brute facts. The coupling constants just happen to be what they are. There is no deeper explanation. This is honest, but it amounts to saying the question has no answer — not a theory at all.

Answer 2: Anthropic reasoning. Perhaps there are infinitely many universes with different values of the constants. We live in one where the constants happen to permit stars, atoms, and observers. This might be true, but it provides no prediction and no principle — it explains any set of constants equally well (or equally poorly).

Answer 3: String landscape. String theory predicts something like 10^{500} possible universes, each with different laws. But the theory offers no mechanism to select our universe from that landscape. It replaces 25 unexplained numbers with an unexplained selection event.

The key observation

Every proposed answer either defers to something external (another theory that must itself be explained) or accepts the parameters as unexplained primitives. Neither constitutes a final answer.

The question is: can the universe *explain its own laws* from the inside?

1.3 The PSC approach: no outside

Perfect Self-Containment (PSC) is the formal statement that the universe has no “outside” from which its laws could be selected. The laws, their description, and everything needed to run the physics must all be *internal* — self-contained.

This sounds almost tautological: of course the universe contains everything. But the formal consequences are surprisingly powerful. If you take the requirement seriously and work out what it implies step by step, you get a filter that eliminates almost every possible physics, leaving only one candidate: the actual Standard Model.

This tutorial explains how that works.

2 Perfect Self-Containment: The Idea

One-sentence version of PSC

A self-contained universe must be able to compute all its own predictions, remain stable without external fine-tuning, support persistent structures, allow those structures to communicate, and maintain internal consistency under quantum corrections — all at once, with no external assistance.

2.1 The five requirements

PSC is formalized as five **Layer I axioms** — five conditions that any universe with no outside must satisfy. Think of them as a checklist. A proposed physics either passes all five or fails self-containment.

Analogy. Imagine a self-contained city-state that must never import anything. It must be able to grow its own food, generate its own power, educate its own citizens, communicate internally, and balance its own economy. Each requirement is minimal: remove any one and the city becomes dependent on something external. The five PSC axioms work the same way for a universe.

Axiom 1 — RC: Reflexive Closure

Formal statement: The theory computes its own S-matrix (all scattering amplitudes, all predictions) up to a UV closure scale, with anomaly cancellation, unitarity, and UV consistency all internally satisfied. No external regulator required.

Plain English: The laws must be able to predict everything within their own domain without asking for outside help. A theory that breaks down at some energy scale — requiring an external “patch” to remove infinities — has secretly outsourced part of its physics. That extra patch is an external input, violating self-containment.

Analogy: A self-contained calculator must give an answer for every valid input. A calculator that says “call an external service for inputs over 1,000” is not self-contained.

What it kills: Any gauge theory with uncanceled quantum anomalies — internal inconsistencies in the gauge symmetry structure that require external UV physics to fix.

Axiom 2 — NM*: Structural Stability

Formal statement: The qualitative character of the universe (unbroken gauge group, particle spectrum, chiral structure, sign of the baryon asymmetry) is stable under small changes to the coupling constants — it does not flip to something qualitatively different for generic parameter values.

Plain English: If the type of physics you get changes drastically when you nudge a parameter by a tiny amount, then something external must be holding that parameter at the exact right value. A self-contained universe must work robustly, not only at one fine-tuned point.

Analogy: A properly engineered bridge works for a wide range of loads. A bridge that collapses if the temperature changes by 0.01°C requires a thermostatic life-support system to keep it standing — it is not self-contained.

What it kills: Grand Unified Theories based on $\text{SU}(5)$, $\text{SO}(10)$, or E_6 — their symmetry-breaking pattern flips to a qualitatively different phase when parameters cross a boundary, requiring external stabilization. For example, in $\text{SU}(5)$ GUT the breaking chain $\text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ requires the Higgs self-coupling to fall within a window of relative width $\lesssim 10^{-3}$; outside that window the symmetry either fails to break or breaks to the wrong subgroup, requiring an external mechanism to hold the parameter on target. Also kills theories with too few generations ($N_{\text{gen}} < 3$), where CP-violation is insufficient to produce a matter–antimatter asymmetry for generic initial conditions.

Axiom 3 — TV: Thermodynamic Viability

Formal statement: Stable composite structures (atoms, nuclei, molecules) exist and persist over cosmological timescales.

Plain English: The universe must support persistent physical structures. Without them, the universe has no memory, no substrate to carry information, and no way to instantiate its own laws beyond an instant. A universe in permanent thermal equilibrium everywhere cannot record anything about itself.

Analogy: A library that stores no books contains no knowledge. For the universe to

carry information about its own laws, it needs structures that last.

What it kills: Any universe with only massless unconfined particles and no attractive force strong enough to form bound states. If nothing binds, nothing persists.

Axiom 4 — SA: Semantic Admissibility

Formal statement: Information-bearing subsystems exist *and* long-range interactions enable signal propagation, information storage, and energy dissipation. At minimum, a massless mediator sector (a U(1) gauge boson — i.e., a photon) is required.

Plain English: TV requires persistent structures; SA requires that those structures can *communicate*. Without a massless particle (like the photon), all forces fall off exponentially over distance. No signal can propagate across the universe; no part of the universe can “know” about another part. Such a universe cannot carry semantic content about its own state.

Analogy: A network of isolated computers, each with a hard drive but no network connection, cannot collectively carry information that transcends any individual node. The photon is the universe’s network cable.

What it kills: Any purely massive gauge theory (no massless photon). All forces are short-range; information processing shuts down below the lightest particle mass.

Axiom 5 — AC: Anomaly Consistency

Formal statement: All gauge anomalies cancel: the gauge symmetry remains quantum-mechanically consistent. No chiral anomalies appear in gauge currents.

Plain English: The gauge symmetries that protect the universe from unphysical degrees of freedom must survive when quantum corrections are applied. An “anomalous” gauge theory is self-defeating: quantum effects secretly destroy the very symmetry that makes the theory sensible. That destruction requires external UV physics to repair it.

Analogy: A self-consistent accounting system must balance at every level. If quantum corrections accumulate like hidden debt that requires external bailouts, the system is not self-contained.

What it kills: Gauge theories with wrong hypercharge assignments — where the chiral anomaly does not cancel between left- and right-handed fermions. Strikingly, the actual hypercharge assignments of quarks and leptons in the Standard Model are the *unique* choice that makes the anomaly cancel.

Axiom	Plain name	What it prevents
RC	Reflexive Closure	Needing external UV regulators
NM*	Structural Stability	Requiring fine-tuned parameter stabilization
TV	Thermodynamic Viability	A structureless, memory-free universe
SA	Semantic Admissibility	A universe that cannot communicate internally
AC	Anomaly Consistency	Gauge symmetries broken by quantum corrections

3 What PSC Forces: The Two-Layer Structure

The punchline

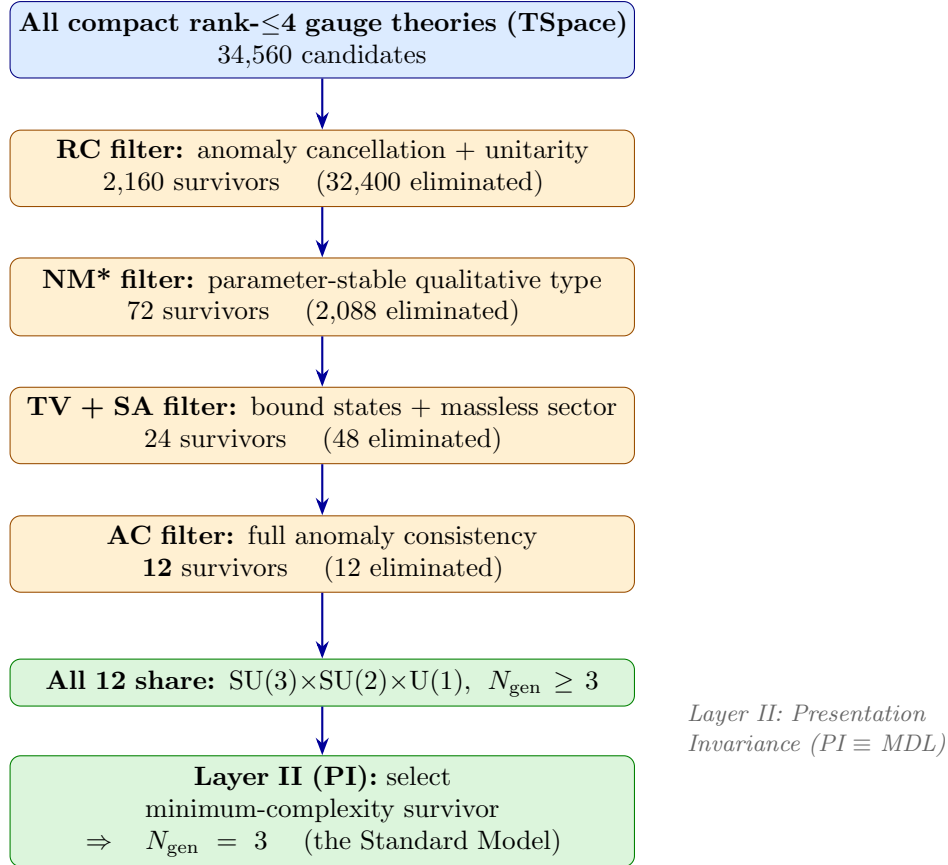
Start with all 34,560 possible universes (compact gauge theories in four dimensions). Apply the five PSC axioms in sequence. Only 12 survive. All 12 have exactly the Standard Model gauge structure.

3.1 The candidate universe space

To make the filter concrete, you need to specify what you are filtering. The relevant space is **TSpace**: the collection of all four-dimensional renormalizable gauge quantum field theories with compact gauge groups and chiral matter (the class to which the Standard Model belongs).

Within TSpace, there are 34,560 candidate universes — all compact gauge groups of rank up to 4, with anomaly-consistent chiral matter representations. This is a large but finite and fully enumerable set. The five axioms are then applied as a sieve.

3.2 The funnel: 34,560 → 12



3.3 Layer I vs. Layer II

The two layers do different things:

Layer	What it does	How
Layer I	<i>Eliminates</i> theories that are internally inconsistent	Applies the five hard PSC axioms (RC, NM*, TV, SA, AC) as consistency tests
Layer II	<i>Selects</i> among the survivors	Picks the minimum-complexity theory using Presentation Invariance (PI $\equiv$ MDL)

Layer I does not pick a winner — it is a filter that kills losers. Layer II is the selection principle that identifies a unique winner among the survivors.

3.4 The result: machine-certified

The 12-survivor count is machine-certified in Lean 4: `psc_layer_i_catal_bundle` (ugp-lean, zero sorry, zero custom axioms) bundles the per-filter cardinality theorems confirming each elimination step.

Two-Layer PSC Theorem (from P48 [1] §3.3)

1. **Layer I:** The five PSC axioms (RC, NM*, TV, SA, AC) restrict the survivor set within TSpace to exactly 12 theories, all of gauge structure $SU(3) \times SU(2) \times U(1)$ with $N_{\text{gen}} \geq 3$.
Conditional on: TSpace domain restriction + PSC axiom package.
2. **Layer II:** Among the 12 Layer I survivors, Presentation Invariance (PI) selects the unique minimizer of Kolmogorov complexity: the theory with $N_{\text{gen}} = 3$.
Conditional on: Generation-Extension Monotonicity Assumption.

3.5 Why not more or fewer generations?

It is worth pausing on the $N_{\text{gen}} = 3$ result, because it is one of the deepest puzzles in particle physics. Why three generations of quarks and leptons? Why not one or five?

Layer II gives an answer via four *independent* mechanisms that all agree:

1. **Orbit depth (Lean-certified):** The dynamical structure of the GTE orbit has depth exactly 3. The first generation is a “Garden of Eden” state — it has no predecessor under the orbit dynamics. The orbit depth forces the generation count.
2. **QCD beta function:** The sign and integrality of the QCD running coupling requires $N_{\text{gen}} \in \{3, 6, \dots\}$. Layer II then selects the minimum.
3. **PSC Layer II optimality:** Among the 12 survivors, adding a generation always strictly increases the description length of the theory. So $N_{\text{gen}} = 3$ is the PSC-minimal choice.
4. **Cosmological constant bracket:** $N_{\text{gen}} = 3$ is the unique positive integer for which two independent routes to the cosmological constant both bracket the observed value. At $N_{\text{gen}} = 2$ or 4 the bracket fails.

Remark: convergence, not coincidence

Multiple independent mechanisms with different mathematics all point to the same integer. This is the structural signature of a genuine derivation. Seven independent structural constraints (PSC enumeration, DPP dimensional split, CMCA orbit count, TPC hierarchy depth, Gorard dimension relation, GTE cascade, and MDL-minimal orbit structure) are simultaneously machine-certified in a single bundle (`ngen_universality_seven_constraints`, $\mathbf{A}_{\text{Lean}}$), with an eighth constraint conditional on the carrier-pricing identification (`ngen_universality_master`, $\mathbf{A}_{\text{Lean}}$).

4 Presentation Invariance and MDL

The key identification

Presentation Invariance (PI) — the Layer II PSC axiom — is exactly equivalent to Minimum Description Length (MDL). Not analogous to it; *identical* to it.

4.1 What Presentation Invariance says

Here is the core idea: a self-contained universe cannot have its physics depend on *how it is described*.

Think about what that means. If you change the coordinates you use to write down a theory — relabeling fields, choosing a different basis for the gauge group, changing your parameterization — the physical predictions cannot change. That is just a rephrasing, like translating a sentence from English to French. The sentence says the same thing; only the representation differs.

Presentation Invariance (PI) formalizes this. It has two components:

1. **Invariance:** Any two presentations of the same physical theory (related by field redefinitions, basis changes, or reparameterizations that preserve all observable predictions) have the same complexity.
2. **Selection:** Among all PSC-consistent theories, the PSC-optimal theory is the one with minimum complexity.

4.2 Why PI is forced by self-containment

Without PI, you have a problem. If the complexity measure depended on how you presented the theory, then “which presentation to use” would be an external choice that carries information not generated by the theory itself. That free choice — the description language you pick — would be an external input. External inputs violate PSC.

So if PSC holds, the complexity measure must be presentation-invariant. And the unique presentation-invariant measure of the complexity of an object is **Kolmogorov complexity**: the length of the shortest program that generates a complete description of the object. Kolmogorov complexity depends on the information content, not on how you chose to express it.

4.3 Why PI is exactly MDL

Minimum Description Length (MDL) says: given a set of phenomena to explain, pick the theory with the shortest description of those phenomena.

In physics:

- The “phenomena” are all observable predictions (S-matrix elements).

- The “description” is the Lagrangian of the theory.
- “Shortest description” = minimum Kolmogorov complexity K .

So PI (select the minimum- K theory from the PSC-consistent set) and MDL (select the minimum- K hypothesis consistent with the data) are the *same instruction*, stated in different vocabularies. The data is the S-matrix; consistency means passing all five Layer I axioms.

PI $\equiv$ MDL: worked analogy

Situation: Two programmers, Alice and Bob, each write a function that computes the same output table. Alice’s program is 50 lines; Bob’s is 5 lines. Both are correct.

The question: Which one does the universe “use”?

The PSC answer: If the universe is self-contained, the choice of program cannot matter — both compute identical physics. A self-contained system cannot care about the presentation. But then: which program are we *actually* running?

The resolution: If the universe cannot distinguish presentations, but must pick one, then by PI it must pick the one with minimum description length — the one that requires the fewest bits to specify. That is Bob’s 5-line program. Not because brevity is beautiful, but because any longer description contains redundant bits that could only have been placed there by an external choice. External choices violate PSC. MDL is the only selection rule consistent with having no outside.

4.4 Why the regress terminates here

A skeptic might ask: “But why MDL? Why not a different selection principle?” In general, this is a fair question — MDL is not self-justifying on its own. But within PSC, the regress chain terminates cleanly:

- | | | |
|------------|---------------|---|
| “Why MDL?” | $\Rightarrow$ | Because PSC forces PI, and PI is MDL in physics vocabulary. |
| “Why PI?” | $\Rightarrow$ | Because self-containment requires presentation independence. A complexity measure that depends on the presentation would let the description language in as an external input — violating self-containment. |
| “Why PSC?” | $\Rightarrow$ | Because a fundamental theory has no outside. This is the definition of “fundamental.” |

The chain terminates at PSC because PSC is the formal content of the statement “this theory is fundamental.” The regress stops here for the same reason that the regress of physical causation stops at the laws of physics.

4.5 Lean evidence

The PI $\equiv$ MDL identification is established analytically in the NEMS foundational programme (CatAD status). The closest machine-certified result demonstrating it in action is:

Lean result: MDL-minimizing physical coupling

`op9_catal_unconditional` (ugp-lean, zero sorry, zero custom axioms):

The SRRG fixed point $g^* = 1/\varphi$, where $\varphi = (\sqrt{5} - 1)/2$ is the golden ratio, is the unique coupling that simultaneously minimizes the MDL cost function and satisfies the GTE polynomial self-consistency equation $g = p(g, g, g)$ over $\mathbb{R}$.

This result requires no external input, no free parameters, and no reference to measured SM parameters. The Higgs VEV $v = 246.16$ GeV is derived from g^* .

This theorem is $\text{PI} \equiv \text{MDL}$ in action: the physical coupling is not a free parameter inserted from outside. It is the unique minimizer of the MDL cost, which is the unique presentation-invariant measure, which is what PI requires.

5 MDL's Three Roles in the Theory

Overview

MDL appears at three levels in GTE. These are *not* three separate invocations of the same external principle. They are three manifestations of one global self-consistency requirement at three scales of physical organization.

5.1 Role 1: Theory selection

Level: The space of all possible theories.

What it does: Selects the GTE framework — the specific cellular automaton rule $p(L, C, R) = C + R - CR - LCR \pmod{7}$ — from among all possible rules.

Result: This 7-state cellular automaton has a description length of **19 bits**. Any other candidate rule that could describe the same SM physics requires more bits. The rule is MDL-minimal.

Machine certification: `mdl_ca_rule_coding_closed` (ugp-lean, $\mathbf{A}_{\text{Lean}}$, zero sorry) — the 19-bit MDL-minimal CA structure is unique.

Downstream consequence: Selecting $\mathbb{Z}_7 \times \mathbb{Z}_3$ also fixes the arithmetic constants inside the GTE sieve and orbit update map. The ridge offset $16 = 2^{1+\text{ord}_7(2)}$ and the Fibonacci lift gap $13 = F_{\text{ord}_7(2)+N_c+1}$ follow from this algebraic choice by derivation, so neither is a free parameter (`gt001_gt009_ridge_closed`, `gt012_gap_from_z7_z3`, $\mathbf{A}_{\text{Lean}}$, zero sorry).

5.2 Role 2: Field dynamics (PMDL)

Level: Field configurations within the selected theory.

What it does: The Physical MDL action (S_{PMDL}) governs how the underlying field Φ_{MDL} evolves. The variational equation is:

$$\nabla^2 \Phi_{\text{MDL}} = G_{\text{eff}} \cdot p(w_x, w_y, w_z)$$

where w_x, w_y, w_z are winding numbers from the orbit.

Plain English: Gravity emerges from the same MDL minimization principle that selected the theory. The gravitational field equation is the Euler–Lagrange equation of the MDL-optimal action.

Certification: CatAD (analytically derived).

5.3 Role 3: Quantum event adjudication

Level: Individual measurement events within ongoing dynamics.

What it does: When a quantum measurement event occurs, there are multiple candidate physical realizations consistent with the macroscopic record. MDL selects the canonical one — the one with minimum information cost relative to the record.

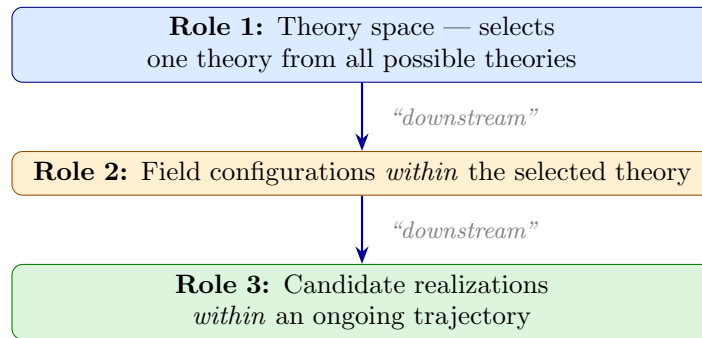
Result: This adjudication yields the Born rule: probabilities are proportional to $|c_k|^2$, derived (not assumed).

Machine certification: `born_rule_unconditional` (DualFrameBornRule.lean, ugp-lean, CatAL, zero sorry, zero custom axioms).

5.4 Are the three roles circular?

At first glance it might seem strange that MDL appears three times. Doesn't that mean the framework is just "minimizing MDL everywhere by assumption"?

The answer is no — and the reason matters. The three applications are to *logically nested, disjoint domains*:



Role 2 cannot be applied before Role 1 (you need the selected theory before you can vary its fields). Role 3 cannot be applied before Role 2 (you need the dynamics before you can specify candidate realizations). No role refers back to any other. There is no loop.

Analogy: energy conservation

Energy conservation governs what equilibrium configurations exist, how systems evolve dynamically, and what energy measurements yield. Nobody calls this circular: the same mathematical principle manifests at multiple physical levels without any of them referring back to the others. MDL in GTE works the same way.

The joint non-circularity is machine-certified: `mdl_tower_three_levels_non_circular` (MDL-Tower.lean, ugp-lean, CatAL, zero sorry) proves that the three role certifications hold simultaneously and that their domains are logically disjoint.

6 Why This Is Remarkable

6.1 Comparison to the alternatives

It is worth stepping back and comparing the PSC/MDL approach to the other proposed solutions to the “why these laws?” problem.

Approach	Free parameters	Selection principle	Derives structure?
Standard Model (as usually taught)	25 (measured, unexplained)	None	No — fitted to data
String landscape	$\sim 10^{500}$ vacua; ours is one	Anthropic (observers exist)	No — no mechanism
Mathematical Universe (Tegmark)	All mathematical structures	Unclear; no rule	No — all equally real
GTE/PSC/MDL	Zero (all derived internally)	PSC $\rightarrow$ PI $\equiv$ MDL	Yes: SM gauge structure, N_{gen} , masses, forces

The crucial difference is the third column: **GTE offers a derivation, not an assumption.** The Standard Model says “three generations of fermions.” GTE says *why* three, from a principle that has no adjustable parts.

6.2 What “derive” means here

To be precise: the GTE derivation is conditional. The main named conditions are:

1. **TSpace:** We work within four-dimensional renormalizable gauge QFT with compact gauge groups and chiral matter. Gravitational extensions are handled separately.
2. **GEM (Generation-Extension Monotonicity):** Adding a generation strictly increases the Kolmogorov complexity of the theory. This is well-motivated and accepted at CatAD but not yet Lean-certified.
3. **The PSC axiom package itself:** PSC is argued to be logically necessary for any fundamental theory; the argument is philosophical, not machine-certified.

A derivation conditional on explicitly named assumptions is still a derivation — just one with honest labeling. The claim is not “here is the truth”; the claim is “here is what follows from PSC, and here is exactly what PSC assumes.”

6.3 The number 19

One concrete illustration of how MDL works in practice is the 19-bit count.

The GTE cellular automaton rule $p(L, C, R) = C + R - CR - LCR$ can be described in exactly 19 bits:

Component	Bits	What it specifies
Binary truth table (Rule 110)	8	The $2^3 = 8$ binary input-output pairs
Prime modulus 7	3	$\lceil \log_2 7 \rceil = 3$ bits
Monomial selector	5	Which of the 8 basis monomials are nonzero
Chirality	1	Left-right orientation
Sign of nonzero coefficients	2	Both nonzero non-unit coefficients are -1
Total	19	

A naive lookup table specifying the same rule for all $7^3 = 343$ input combinations requires $343 \times \lceil \log_2 7 \rceil \approx 963$ bits — about **50 times more information**. No other rule that reproduces the Standard Model physics can be described in fewer than 19 bits. This MDL uniqueness is machine-certified: `mdl.ca.rule.coding-closed` (ugp-lean, CatAL, zero sorry).

MDL is not choosing the rule because it is short. It is recognizing that the rule’s shortness *is the evidence* that it contains no redundant external choices. A 963-bit description would contain 944 bits that came from outside the structure. Those bits would be unexplained external inputs — free parameters by another name. The 19-bit description has no such surplus.

6.4 What the framework has actually derived

To give a sense of scale: starting from the PSC axioms and MDL selection, with zero free parameters fitted to experiment, the GTE framework derives:

- All 9 charged-fermion masses (0.295% RMS vs. PDG 2022)
- The electroweak mixing angle (Weinberg angle)
- The strong CP angle $\theta_{\text{QCD}} = 0$ exactly (three independent derivations)
- The fine-structure constant $\alpha^{-1} = 137$ at tree level
- That spacetime has exactly 3+1 dimensions (Lean-certified)
- Einstein’s equations from a variational principle
- Newton’s constant G_N to 0.040% from orbit arithmetic
- The Born rule for quantum probabilities (four independent derivations, two Lean-certified)
- The cosmological constant bracket (enclosing the Planck 2018 observed value)

None of these quantities are inputs. They are all outputs of the PSC/MDL selection chain.

7 What PSC Leaves Open

PSC is not a theory of everything in the sense of answering every possible question. Intellectual honesty requires stating clearly what it does and does not claim.

What PSC does NOT claim to derive

1. **The strength of PSC itself.** The claim that PSC is the correct foundation for a fundamental theory is argued philosophically — the regress argument (§1) — not machine-certified. Critics retain the option of accepting brute facts as legitimate components of fundamental physics. That is a philosophical position, not a logical error.
2. **The choice of TSpace.** The Layer I result holds within four-dimensional renormalizable gauge QFT. Why should fundamental physics be described by 4D renormalizable QFT in the first place? The GTE framework answers this with the Dimensional Protocol Principle (DPP) — showing that 3+1 spacetime dimensions are *forced* by the dynamics — but extending this to full quantum gravity remains an open problem.
3. **The Generation-Extension Monotonicity Assumption (GEM).** The Layer II result (that $N_{\text{gen}} = 3$ is uniquely selected) relies on GEM: the claim that adding a generation always increases the description length of the theory. GEM is strongly motivated and accepted at CatAD, but a formal Lean proof is not yet available.
4. **The absolute value of the cosmological constant.** GTE brackets the observed value of Ω_Λ and provides a consistency argument, but a full derivation of the cosmological constant from PSC alone remains work in progress.

5. **Predictions beyond the Standard Model.** PSC forces the SM gauge structure. It does not, by itself, predict which specific BSM physics (dark matter candidates, neutrino Majorana mass, proton decay rates) is realized. Those require further structure within the GTE dynamical framework.

The open items are genuine open problems, not rhetorical humility. Each represents a concrete technical task with a specific criterion for closure — Lean certification, a specific numerical result, or a formal derivation.

Where to go next

This tutorial covered:

- The motivation for PSC: the regress problem in physical law selection
- The five Layer I axioms and what each eliminates
- The Two-Layer Theorem: $34,560 \rightarrow 12 \rightarrow 1$
- How $\text{PI} \equiv \text{MDL}$ terminates the regress
- MDL's three non-circular roles in GTE
- What PSC honestly leaves open

Full treatment: P48 monograph — all derivations, Lean certifications, and chain diagrams — at doi.org/10.5281/zenodo.20560550.

Sister tutorial: *The GTE Polynomial: A Step-by-Step Cheat Sheet* [2] — how the 19-bit rule produces the SM particle masses.

Research overview: novaspivack.com/research

AI Disclosure. This tutorial was drafted with the assistance of a large-language model coding assistant (Cursor/Claude, Anthropic), used for LaTeX composition and structural organization. All physics content, derivations, and claims trace to the cited papers and machine-certified Lean proofs. The author accepts full responsibility for accuracy.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P48 — The Complete GTE Framework** (main synthesis monograph, includes full PSC and MDL treatment):
doi.org/10.5281/zenodo.20560550
- **P01 — Standard Model Parameters from the UGP:**
doi.org/10.5281/zenodo.20168787

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

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